

Supplementary material: Continuous multisensory proprioception model

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1 Methods

1.1 General equations for LQG control

The state of the environment $\mathbf{x}_t$ evolves as a function of the previous state and the action $\mathbf{u}_t$ performed by the agent according to a discrete-time linear dynamical system with Gaussian noise

$$\mathbf{x}_{t+1} \sim \mathcal{N}(\mathbf{A}\mathbf{x}_t + \mathbf{B}\mathbf{u}_t, \mathbf{\Sigma}_x), \quad (1)$$

with noise covariance $\mathbf{\Sigma}_x$.

At each time step, the agent receives a noisy observation $\mathbf{y}_t$ of the state,

$$\mathbf{y}_t \sim \mathcal{N}(\mathbf{H}\mathbf{x}_t, \mathbf{\Sigma}_y), \quad (2)$$

which is a linearly transformed version of the state with additive Gaussian noise with covariance $\mathbf{\Sigma}_y$.

The cost function is quadratic in the state and the actions

$$\sum_{t=1}^T \mathbf{x}_t^\top \mathbf{Q} \mathbf{x}_t + \mathbf{u}_t^\top \mathbf{R} \mathbf{u}_t. \quad (3)$$

This control problem is called linear-quadratic-Gaussian (LQG) control and has a well-known optimal solution, which is composed of optimal state estimation using the Kalman filter (KF: Kalman, 1960a) and optimal feedback control using the linear-quadratic regulator (LQR: Kalman, 1960b)

The KF iteratively computes a posterior distribution about the state given a sequence of sensory observations $p(\mathbf{x}_t | \mathbf{y}_1, \dots, \mathbf{y}_t) = \mathcal{N}(\hat{\mathbf{x}}_t, \mathbf{\Sigma}_t)$, where the mean state estimate and its covariance are updated according to

$$\hat{\mathbf{x}}_{t+1} = \mathbf{A}\hat{\mathbf{x}}_t + \mathbf{B}\mathbf{u}_t + \mathbf{K}_{t+1}(\mathbf{y}_{t+1} - \mathbf{H}(\mathbf{A}\hat{\mathbf{x}}_t + \mathbf{B}\mathbf{u}_t)) \quad (4)$$

$$\mathbf{\Sigma}_{t+1} = (\mathbf{I} - \mathbf{K}_{t+1}\mathbf{H})\mathbf{P}_{t+1}, \quad (5)$$

where $\mathbf{K}_t$ is defined as

$$\mathbf{K}_{t+1} = \mathbf{P}_{t+1} \mathbf{H}^\top (\mathbf{H} \mathbf{P}_{t+1} \mathbf{H}^\top + \boldsymbol{\Sigma}_y)^{-1}, \quad (6)$$

$$\mathbf{P}_{t+1} = \mathbf{A} \boldsymbol{\Sigma}_t \mathbf{A} + \boldsymbol{\Sigma}_x. \quad (7)$$

$$(8)$$

The linear-quadratic regulator defines an policy as a function of the current state estimate

$$\mathbf{u}_t = \mathbf{L}_t \hat{\mathbf{x}}_t, \quad (9)$$

which minimizes the expected cost by solving a dynamic programming problem. For a derivation of $\mathbf{L}_t$ and a more detailed derivation of $\mathbf{K}_t$, see e.g. Kochenderfer et al. (2022).

1.2 Target and hand dynamics

We consider two different models of the subject’s hand dynamics, which we call the velocity control model and the point mass model. In both models, the target follows a random walk with variance σ_{rw}^2 and the subject controls the left hand to track the target.

1.2.1 Velocity control model

In the velocity control model, the latent state contains the target position and the subject’s hand position,

$$\mathbf{x}_t = \begin{bmatrix} x_t^{\text{right}} \\ x_t^{\text{left}} \end{bmatrix}, \quad (10)$$

where the target follows a random walk and the subject controls the velocity of the left hand. The generative dynamics are

$$\mathbf{x}_{t+1} = \mathbf{A} \mathbf{x}_t + \mathbf{B} \mathbf{u}_t + \mathbf{v}_t, \quad \mathbf{v}_t \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_x), \quad (11)$$

$$\mathbf{A} = \mathbf{I}, \quad \mathbf{B} = dt \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \boldsymbol{\Sigma}_x = \begin{bmatrix} \sigma_{\text{rw}}^2 & 0 \\ 0 & \sigma_{\text{control}}^2 \end{bmatrix}. \quad (12)$$

The state cost penalizes the tracking error,

$$\mathbf{Q} = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, \quad \mathbf{R} = [c], \quad (13)$$

where c is a free parameter representing the scaling of the action cost.

1.2.2 Point mass model

In the point mass model, the subject’s hand follows a point-mass dynamical system with position, velocity, and force states, while the target still follows a random walk. In the code, this model is used in the form

$$\mathbf{x}_t = \begin{bmatrix} x_t^{\text{right}} \\ x_t^{\text{left}} \\ v_t^{\text{left}} \\ f_t^{\text{left}} \end{bmatrix}, \quad (14)$$

with the left-hand component obtained from a continuous-time point-mass system commonly used to model reaching movements (see e.g. Kasuga et al., 2022, for details). Concretely, the continuous-time dynamics for the hand’s position, velocity and muscle force are

$$m\dot{v}^{\text{left}} = -gv^{\text{left}} + f^{\text{left}}, \quad (15)$$

$$\tau\dot{f}^{\text{left}} = -f^{\text{left}} + u, \quad (16)$$

where g is a damping constant, m the mass, τ the muscle time constant and u the motor command. Writing these equations together with $\dot{x}^{\text{left}} = v^{\text{left}}$ yields a linear continuous-time system with matrices $\mathbf{A}_c$ and $\mathbf{B}_c$. Discretizing this linear system with time step dt (using matrix exponential or standard zero-order hold discretization) yields the discrete-time matrices $\mathbf{A}_{\text{pm}}$ and $\mathbf{B}_{\text{pm}}$. The continuous-time process noise is transformed to the discrete-time covariance Σ_{pm} via Van Loan’s method (Van Loan, 2003).

The full discrete-time dynamics for the whole state are therefore

$$\mathbf{A} = \text{diag}(\mathbf{I}_1, \mathbf{A}_{\text{pm}}), \quad (17)$$

$$\mathbf{B} = \text{diag}(0, \mathbf{B}_{\text{pm}}), \quad (18)$$

$$\Sigma_{\mathbf{x}} = \text{diag}(\sigma_{\text{rw}}^2, \Sigma_{\text{pm}}). \quad (19)$$

The cost function for the point mass model penalize the squared error between the response and target positions and the squared control signal using the state and action cost matrices

$$\mathbf{Q} = \text{diag}\left(500 \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, 0, 0\right), \quad (20)$$

$$\mathbf{R} = [c], \quad (21)$$

where again c is a free parameter representing the scaling of the action cost. The arbitrary rescaling by a factor of 500 in $\mathbf{Q}$ was empirically chosen such that the scaling of the parameter c is in a similar range compared to the velocity control model.

1.2.3 Delayed multisensory observation model

We model delays by augmenting the state vector (cf. Izawa & Shadmehr, 2008) and multisensory observations by assuming independent noisy sensory channels

for the different sensory modalities (cf. Crevecœur et al., 2016; Kasuga et al., 2022).

Each sensory observation is modelled as the signed difference between the right and left hand positions. Each sensory modality $k \in \{\text{prop}, \text{vis}\}$ has its own delay δ_k and its own noise variance σ_k^2 :

$$\mathbf{y}_t^{(k)} = \mathbf{H}\mathbf{x}_{t-\delta_k} + \mathbf{w}_t^{(k)}, \quad \mathbf{w}_t^{(k)} \sim \mathcal{N}(\mathbf{0}, \Sigma_{\mathbf{y}}^{(k)}), \quad (22)$$

$$\mathbf{H} = \begin{bmatrix} 1 & -1 & 0 & \dots \end{bmatrix}, \quad \Sigma_{\mathbf{y}}^{(k)} = [\sigma_k^2]. \quad (23)$$

The complete sensory signal received by the agent is

$$\mathbf{y}_t = [\mathbf{y}_t^{\text{prop}}, \mathbf{y}_t^{\text{vis}}]^\top. \quad (24)$$

We set $\delta_{\text{vis}} = 2$ and $\delta_{\text{prop}} = 1$, which at our frame rate of $dt = 0.075$ s corresponds to 150 ms and 75 ms sensory delays for vision and proprioception, respectively. The noise variances are free parameters, which are estimated from the data.

To handle sensory delays, we augment the latent state by stacking previous states up to the maximum delay $\delta_{\text{max}} = \max_k \delta_k$ and adjust the cost and dynamics matrices following Izawa and Shadmehr (2008) such that the previous δ_{max} states are stored in the state vector, but do not affect the dynamics or cost function:

$$\tilde{\mathbf{x}}_t = [\mathbf{x}_t, \mathbf{x}_{t-1}, \dots, \mathbf{x}_{t-\delta_{\text{max}}}]^\top, \quad (25)$$

$$\tilde{\mathbf{x}}_{t+1} = \tilde{\mathbf{A}}\tilde{\mathbf{x}}_t + \tilde{\mathbf{B}}\mathbf{u}_t + \tilde{\mathbf{v}}_t, \quad (26)$$

$$\tilde{\mathbf{Q}} = \text{blkdiag}(\mathbf{Q}, \mathbf{0}, \dots, \mathbf{0}), \quad (27)$$

$$\tilde{\mathbf{A}} = \begin{bmatrix} \mathbf{A} & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{I} & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & \mathbf{I} & \dots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \dots & \mathbf{I} \end{bmatrix}. \quad (28)$$

Each sensory modality k then uses an augmented observation matrix $\tilde{\mathbf{H}}^{(k)}$ that picks out the appropriately delayed state component from $\tilde{\mathbf{x}}_t$ (Crevecœur et al., 2016):

$$\mathbf{y}_t^{(k)} = \tilde{\mathbf{H}}^{(k)}\tilde{\mathbf{x}}_t + \mathbf{w}_t^{(k)}, \quad (29)$$

$$\tilde{\mathbf{H}}^{(k)} = [\mathbf{0} \quad \dots \quad \mathbf{0} \quad \mathbf{H} \quad \mathbf{0} \quad \dots \quad \mathbf{0}], \quad (30)$$

where the block $\mathbf{H}$ appears in the column block corresponding to delay δ_k and all other blocks are zero.

1.2.4 Model with visual offset

To model a visual offset, we augment the latent state with a scalar bias variable b_t that shifts the visual observation but not the proprioceptive observation. If

$\mathbf{x}_t^{\text{base}}$ denotes the original latent state for either model, the augmented state is

$$\mathbf{x}_t = \begin{bmatrix} \mathbf{x}_t^{\text{base}} \\ b_t \end{bmatrix}. \quad (31)$$

The proprioceptive observation matrix is

$$\mathbf{H}_{\text{prop}} = \begin{bmatrix} 1 & -1 & 0 \end{bmatrix}, \quad (32)$$

and the visual observation matrix is

$$\mathbf{H}_{\text{vis}} = \begin{bmatrix} 0 & -1 & 1 \end{bmatrix}. \quad (33)$$

For the point mass model, the corresponding matrices are

$$\mathbf{H}_{\text{prop}} = \begin{bmatrix} 1 & -1 & 0 & 0 & 0 \end{bmatrix}, \quad (34)$$

$$\mathbf{H}_{\text{vis}} = \begin{bmatrix} 0 & -1 & 0 & 0 & 1 \end{bmatrix}. \quad (35)$$

In both cases, the bias state affects only the visual observation.

To simulate behavior in the calibration phase of the experiment, we took the posterior mean parameter estimates for each model and each participant. For each level of visual offset in the calibration phase, ranging from 0 to 5 cm in steps of 0.5cm, we simulated 20 trials of tracking behavior.

1.3 Model fitting

We fit the model to the behavioral data using inverse optimal control (IOC) for LQG system (Straub & Rothkopf, 2022). More concretely, IOC defines the likelihood of a trajectory of observed states conditional on the model parameters $p(\mathbf{x}_{1:T}^{(i)}|\boldsymbol{\theta})$, where $\mathbf{x}_{1:T}^{(i)}$ is the sequence containing T time steps of observed left and right hand positions in trial i . For details about how the likelihood of an individual trial is computed, see Straub and Rothkopf (2022). We assume that the individual trials are independent realizations of tracking behavior generated from the optimal control model with the same parameters $\boldsymbol{\theta}$, s.t. the likelihood of a dataset $\mathcal{D}$ is given by $p(\mathcal{D}|\boldsymbol{\theta}) = \prod_i p(\mathbf{x}_{1:T}^{(i)}|\boldsymbol{\theta})$.

To model the experiment with unisensory and multisensory conditions, we assume that the same action cost c and action variability σ_{action} applies across all conditions. We further assume that there is one proprioceptive uncertainty parameter σ_{prop} that is active across all conditions. Furthermore, there are two visual uncertainty parameters $\sigma_{\text{vis},1}$ and $\sigma_{\text{vis},2}$, one for each kind of visual stimulus in the experiment.

Overall, the model has five free parameters $\boldsymbol{\theta} = \{c, \sigma_{\text{action}}, \sigma_{\text{prop}}, \sigma_{\text{vis},1}, \sigma_{\text{vis},2}\}$, where $\sigma_{\text{vis},1}$ and $\sigma_{\text{vis},2}$ are the visual noise parameters for the two levels of visual stimuli in the experiment.

We define weakly informative prior distributions over the parameters:

$$\begin{aligned} c &\sim \text{HalfNormal}(1) \\ \sigma_{\text{action}} &\sim \text{HalfNormal}(1) \\ \sigma_{\text{prop}} &\sim \text{HalfNormal}(20) \\ \sigma_{\text{vis},1}, \sigma_{\text{vis},2} &\sim \text{HalfNormal}(20) \end{aligned}$$

To test for optimal integration of visual and proprioceptive information, we consider different versions of the model. They differ only in how the multisensory condition is modeled, while the unisensory (visual and proprioceptive) conditions are always modeled in the same way.

1. In the *optimal integration* model, both visual and proprioceptive information is used in the multisensory condition and they are combined optimally as prescribed by a Kalman filter.
2. In the *proprioceptive dominance* model, only the proprioceptive sensory modality is used in the multisensory condition.
3. In the *visual dominance* model, only the visual sensory modality is used in the multisensory condition.
4. As a heuristic implementation of an *equal integration* model, we assume that averaging happens at the level of parameters, such that both sensory modalities use the same parameter $\sigma_i = \sqrt{(\sigma_{\text{vis},i}^2 + \sigma_{\text{prop}}^2)}/2$ in the multisensory condition, while unisensory conditions still use their own modality-specific parameters $\sigma_{\text{vis},i}$ and σ_{prop} , respectively.

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